

Year 10
Pathway 1 and 2



Baldivis
Secondary College

Mini Test 1
Indices and Algebraic fractions

Name: Answer Key

Score: /44
Total Time: 45 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Calculators may be used. Calculators may be used. 1 side A4 notes (handwritten) allowed.

1 Simplify $3a + 2b - a + 5b$ [1]

$$2a + 7b$$

2 Evaluate 4^5 [1]

$$4 \times 4 \times 4 \times 4 \times 4 = 1,024$$

3 Evaluate 9^0 [1]

$$1$$

4 Simplify $4^5 \times 5$ leaving your answer index form. [1]

$$4^5 \times 5$$

5 Simplify $x^3 \div x^7$ Leaving your answer with positive indices [1]

$$x^{3-7} = x^{-4} = \frac{1}{x^4}$$

6 Simplify $12x^5 \div 3x^2$ [1]

$$4x^3$$

7 Simplify $(y^{-5})^6$ [1]

$$y^{-30}$$

8 Simplify $(3x^4y)^5$ [2]

$$243x^{20}y^5$$

9 Expand $3(2x + 1)$ [1]

$$6x + 3$$

10 Expand $-2x(x - 3)$ [1]

$$-2x^2 + 6x$$

11 Expand $(x+8)(x+1)$

[1]

$$x^2 + 9x + 8$$

12 Expand $(x+3)(x-2)$

[1]

$$x^2 + x - 6$$

13 Factorise $16y + 32xy^3$

[2]

$$16y(1 + 2xy^2)$$

14 Expand $(a-3)^2$

[2]

$$(a-3)(a-3) = a^2 - 6a + 9$$

15 Expand $(3p-2)(5p+11)$

[2]

$$15p^2 + 23p - 22$$

16 Expand $(3x+3)^2 - 2(x+1)$

[2]

$$\begin{aligned} & (3x+3)(3x+3) - 2(x+1) \\ &= 9x^2 + 18x + 9 - 2x - 2 \\ &= 9x^2 + 16x + 7 \end{aligned}$$

17 Simplify and leave your answer in index form

[1]

$$\begin{aligned} & \frac{2^5 \times 3^2 \times 5^6}{5^2 \times 2^3} \\ &= \frac{2^5 \times 3^2 \times 5^6}{2^3 \times 5^2} \\ &= 2^2 \times 3^2 \times 5^4 \end{aligned}$$

18 Simplify and leave your answer in index form

[3]

$$\begin{aligned} & \frac{(3^2 \times 2^6)^3}{2^{11} \times 3} \\ &= \frac{3^6 \times 2^{18}}{3^1 \times 2^{11}} = 3^5 \times 2^7 \end{aligned}$$

19 Simplify and leave your answer in index form

[2]

$$\begin{aligned} & \frac{x+3}{8} + \frac{2x}{4} \\ &= \frac{x+3 + 2(2x)}{8} = \frac{x+3+4x}{8} \\ &= \frac{5x+3}{8} \end{aligned}$$

20 Simplify and leave your answer in index form

[3]

$$\begin{aligned} & \left(\frac{5}{8^2}\right)^3 \times \left(\frac{4^2}{5}\right)^4 = \frac{5^3}{8^6} \times \frac{4^8}{5^4} \\ &= \frac{5^3}{(2^3)^6} \times \frac{(2^2)^8}{5^4} \\ &= \frac{5^3}{2^{18}} \times \frac{2^{16}}{5^4} = \frac{5^{-1} \cdot 2^{-2}}{5 \cdot 2^2} = \frac{1}{5} \cdot \frac{1}{2^2} \end{aligned}$$

21 Simplify and leave your answer in index form [2]

$$7p^2q^5 \div 56p^6q^8$$

$$= \frac{1 \cancel{7} p^2 \cancel{q^5}}{8 \cancel{56} p^6 \cancel{q^8}} = \frac{p^{-4}}{8}$$

or $\frac{1}{8p^4}$

23 Simplify

$$\frac{2x+2}{5} + \frac{x-2}{5}$$

$$= \frac{(2x) + (2) + (x) - (2)}{5} = \frac{3x}{5}$$

25 The product of x and y is added to the reciprocal of the sum of x and y .

(a) Write down the expression that describes this information [1]

$$xy + \frac{1}{x+y} \text{ or } \frac{1}{x+y} + x \cdot y$$

(b) Write the expression in part (a) with a common denominator. [3]

$$\frac{1}{x+y} + \frac{x \cdot y}{1} = \frac{1 + (x \cdot y)(x+y)}{x+y} = \frac{1 + x^2y + xy^2}{x+y}$$

22. Simplify and leave your answer in index form [1]

$$\frac{x}{2} + \frac{4x}{2} = \frac{5x}{2}$$

24 Simplify

$$\frac{c}{3} \times \frac{5d}{7} = \frac{5cd}{21}$$

26 Simplify

$$\frac{3m}{7} + \frac{m}{3} = \frac{3(3m) + 7(m)}{21}$$

$$= \frac{9m + 7m}{21}$$

$$= \frac{16m}{21}$$

End of Test

